

CUET PG MCA (BOOLEAN ALGEBRA PYQS)

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Topic Included: Boolean Algebra, Logic Gates, Karnaugh Maps, and Combinational Circuits

Question 1

CUET PG 2021 | Operator Precedence

In the expression $Y + X'$, which operator will be evaluated first?

- (a) ' (complement)
- (b) + (OR)
- (c) . (AND)
- (d) , (comma)

Question 2

CUET PG 2021 | Boolean Algebra Laws

Which of the following Boolean laws/identities is false?

- (a) $x + y = y + x$
- (b) $x \cdot y = y \cdot x$
- (c) $x \cdot x' = 1$
- (d) $x + x' = 1$

Question 3

CUET PG 2021 | Expression Simplification

A Boolean function $x'y' + xy + x'y$ is equivalent to:

- (a) $x' + y'$
- (b) $x + y$
- (c) $x + y'$
- (d) $x' + y$

Question 4

CUET PG 2021 | Logic Gates & Circuits

Which of the following logic operations is performed by the given combinational circuit?

- (a) Exclusive-OR
- (b) Exclusive-NOR
- (c) NAND
- (d) NOR

Question 5

CUET PG 2021 | Logic Gates

The output of an AND gate with three inputs A, B and C is HIGH when:

- (a) $A = 1, B = 1, C = 0$
- (b) $A = 0, B = 0, C = 0$
- (c) $A = 1, B = 1, C = 1$
- (d) $A = 1, B = 0, C = 1$

Question 6

CUET PG 2021 | Karnaugh Map

Don't care conditions can be used for simplifying Boolean expressions in:

- (a) registers
- (b) terms
- (c) K-maps
- (d) latches

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Question 7

CUET PG 2021 | Boolean Minterms

In the expression $A + BC$, the total numbers of minterms will be:

- (a) 2
- (b) 3
- (c) 4
- (d) 5

Question 8

CUET PG 2022 | Boolean Logic

Given truth table AB: 00→1, 01→0, 10→0, 11→0. Which Boolean function does it represent?

- (a) OR
- (b) XOR
- (c) NOR
- (d) XNOR

Question 9

CUET PG 2022 | Expression Simplification

The simplified form of Boolean expression $AB + AB'$ is:

- (a) A
- (b) B
- (c) $1 + A$
- (d) $1 + B'$

Question 10

CUET PG 2022 | Boolean Identities

Match List I with List II for Boolean identities:

List I:

- (A) $(x + y)'$
- (B) $x + 1$
- (C) $(xy)'$
- (D) $x + 0$

List II:

- (I) 1
- (II) $x' + y'$
- (III) $x' \cdot y'$
- (IV) x

- (a) A-II, B-I, C-III, D-IV
- (b) A-II, B-IV, C-I, D-III
- (c) A-III, B-I, C-IV, D-II
- (d) A-III, B-I, C-II, D-IV

Question 11

CUET PG 2022 | Boolean Algebra

The simplified form of Boolean Expression $A'BC + ABC$ is:

- (a) AC
- (b) AB
- (c) BC
- (d) 0

Question 12

CUET PG 2022 | XOR & XNOR

If (+) and (-) denote exclusive-OR and exclusive-NOR operations, which of the following is not correct?

- (a) $(P (+) Q)' = P (-) Q$
- (b) $P' (+) Q = P (-) Q$
- (c) $P' (+) Q' = P' (+) Q$
- (d) $(P (+) P) + Q = (P (-) P) (-) Q$

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Question 13

CUET PG 2023 | Boolean Algebra

If x, w, y, z are Boolean variables, which of the following is incorrect?

- (a) $wx + w(x+y) + x(x+y) = x + wy$ (b) $w x'(y + z') + w'x = w' + x + y'z$
(c) $(w x'(y + x z') + w' x') y = x y'$ (d) $(w + y)(wxy + wyz) = wxy + wyz$

Question 14

CUET PG 2023 | Boolean Function

The number of possible Boolean functions that can be defined for n Boolean variables over n -valued Boolean algebra is:

- (a) $n^{(2^n)}$ (b) $2^{(n^2)}$
(c) $2^{(2^n)}$ (d) $n^{(n^n)}$

Question 15

CUET PG 2023 | K-Map Simplification

The function represented by the k-map given in 2023 paper is:

- (a) $BC + AB + AC'$ (b) $BC' + AB + AC'$
(c) $(B (+) C)'$ (d) $A \cdot BC$

Question 16

CUET PG 2025 | Boolean Laws & Identities

Which of the following Boolean expression is/are true?

- (A) $A + AB = A$,
(B) $(A+B)' = A'B'$,
(C) $(A')' = A$,
(D) $(AB)' = A' + B'$
(a) A, B and D only (b) A and D only
(c) A, B, C and D (d) B and D only

Question 17

CUET PG 2025 | Logic Gates

The following circuit generates the same output as:

- (a) XOR GATE (b) XNOR GATE
(c) NAND GATE (d) NOR GATE

Question 18

CUET PG 2025 | K-Map

Consider the given Karnaugh Map. Minimal Function generated by this Karnaugh map is:

- (a) $Q'S + P'Q$ (b) $Q'S' + P'S'$
(c) $P'Q' + P'S' + Q'S'$ (d) $PQ + Q'S' + P'S'$

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Question 19

CUET PG 2025 | Boolean Algebra

What should be the output of the boolean expression after simplifying to a minimum number of variables? $a'b' + ab + a'b$

- (a) $a + b'$
- (b) $a' + b$
- (c) $a' + b'$
- (d) $a + b$

Question 20

CUET PG 2026 | De Morgan's Law

The complement of $(A + B)$ is:

- (1) $A' + B'$
- (2) $A \cdot B'$
- (3) $A + B$
- (4) $A' \cdot B'$

Question 21

CUET PG 2026 | K-Map Steps

Arrange the following steps of K-Map simplification in correct order: (A) Write minimized SOP/POS (B) Plot truth table outputs (C) Identify prime implicants (D) Draw groups of adjacent 1s

- (1) A, B, C, D
- (2) B, C, A, D
- (3) C, A, B, D
- (4) B, D, C, A

Question 22

CUET PG 2026 | Universal Gate

Which of the following is a Universal logic gate?

- (1) AND
- (2) OR
- (3) NAND
- (4) XOR

Question 23

CUET PG 2026 | Logic Gates Matching

Match List I (Logic Gates) with List II (Output conditions):

List I: (A) AND, (B) OR, (C) NAND, (D) NOR

List II: (I) Output is 1 only if all inputs are 0, (II) Output is 0 only if all inputs are 1, (III) Output is 1 if at least one input is 1, (IV) Output is 1 only if all inputs are 1.

- (1) A-IV, B-III, C-II, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-II, C-IV, D-I
- (4) A-IV, B-III, C-I, D-II

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Answer Key (2021 – 2026)

Q. No.	Year	Correct Answer	Q. No.	Year	Correct Answer
1	2021	(a) ' (complement)	13	2023	(c)
2	2021	(c) $x \cdot x' = 1$	14	2023	(c) $2^n(2^n)$
3	2021	(d) $x' + y$	15	2023	(b) $BC' + AB + AC'$
4	2021	(a) Exclusive-OR	16	2025	(c) A, B, C and D
5	2021	(c) $A = 1, B = 1, C = 1$	17	2025	(b) XNOR GATE
6	2021	(c) K-maps	18	2025	(b) $Q'S' + P'S'$
7	2021	(d) 5	19	2025	(b) $a' + b$
8	2022	(c) NOR	20	2026	(4) $A' \cdot B'$
9	2022	(a) A	21	2026	(4) B, D, C, A
10	2022	(d) A-III, B-I, C-II, D-IV	22	2026	(3) NAND
11	2022	(c) BC	23	2026	(2) A-I, B-II, C-III, D-IV
12	2022	(c)			